

Ekam Kooner

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Skills

Languages: Rust, C, C++, Python, Java, Javascript/Typescript

Tools: Matlab, Git, Fusion360, Solidworks, Altium, STM32CubeIDE, Spring Boot, RESTful APIs

Embedded Systems: PID control, Odometry, EMG Signal Processing

Experience

Embedded Software Engineer, UBC Bionics

September 2025 – Present

- Aided development of **EMG-controlled** trans-radial prosthetic arm in preparation for CYBATHLON 2028 as a member of the Electrical/Embedded subteam.
- Implemented a complete **Rust-to-Python** interface using PyO3 and maturin, enabling the core Rust codebase to be referenced directly with Python.
- Reduced inter-layer communication latency by **97.5%** from **2ms to 50ns** by migrating from an SGCP protobuf over MPSC channels to direct Python bindings using PyO3.
- Engineered **Rust-based I2C multiplexing** and fault-detection modules for a custom BMS, integrating a **TCA9548A switch** and dual **MAX17049** fuel gauges to enforce strict safety cutoffs across a 2S2P lithium-ion array.

Robotics Software Engineer, WCHS VEX Robotics

April 2023 – May 2025

- Developed **autonomous C++ routines** using **odometry**, **IMU**, and **gyroscope** data to achieve high-precision navigation and reliable pose estimation.
- Implemented and tuned **PID controllers** and holonomic drive algorithms to stabilize drivetrain motion and improve trajectory accuracy.
- Optimized carrot curve pathfinding for autonomous routine using a **custom pure pursuit algorithm**.
- Technical Team Lead for **Alberta's #1 VEX Robotics team**, overseeing 10 members & leading autonomous programming while competing internationally at the VEX World Championship in Dallas, Texas.

Projects

HoneyKey: Honeybot API Security System – *Best Cybersecurity Hack Finalist @ nwHacks*

- Built **FastAPI** backend with **24 REST endpoints**, implementing middleware for request interception and structured telemetry logging.
- Designed **SQLite schema** (4 tables) and optimized queries to transform raw HTTP traffic into incidents
- Developed SOC-style report generation pipeline producing structured outputs (severity, evidence, actions) from aggregated request telemetry.
- Mapped attacker behavior to 5+ **MITRE ATT&CK** techniques using request pattern analysis (enumeration, credential access, scanning) for standardized threat classification.

Barrage: API Load Testing & Performance Monitor

- Built a concurrent API load tester in **Java 17** and **Spring Boot**, using **ExecutorService** thread pools and RESTful endpoints to fire and measure simultaneous requests.
- Implemented a **JUnit 5** test suite for **QA validation**, covering failure handling, thread-safety, and edge cases across concurrent execution scenarios.
- Developed a real-time dashboard in **HTML5**, **CSS**, and **jQuery** visualizing response timelines, status codes, and pass/fail rates against a Spring Boot backend.
- Ensured thread-safe orchestration using **AtomicInteger** and **CountDownLatch**, computing throughput and failure rates across configurable concurrency levels.

Education

University of British Columbia – Bachelor of Applied Science

September 2025 – Present